

The Sedenion Seed: Solving the Early Quasar Mystery via Fractal Algebraic Accretion

Aaron M. Schutz
Independent Researcher

December 14 2025

Abstract

The discovery of super-massive black holes (SMBHs) exceeding $10^9 M_\odot$ at redshifts $z > 7$ presents a fundamental paradox for standard Λ CDM accretion models, which are rigidly constrained by the Eddington limit. We propose a resolution within the Axiomatic Physical Homeostasis (APH) framework. We posit that the primordial Sedenion-to-Octonion phase transition was not spatially uniform but fractal, characterized by a specific Hausdorff dimension $D_F \approx 2$. This transition left behind high-density topological defects—*Sedenion Dust*—which served as *Direct Collapse Seeds* with initial masses of $10^4 - 10^6 M_\odot$, bypassing the stellar-mass bottleneck. We demonstrate that the *Super-Stiff* equation of state of these cores ($\Gamma = \pi$) enables super-Eddington accretion rates by suppressing radiative feedback via internal algebraic convection. We derive the Initial Mass Function (IMF) of these seeds and present falsifiable predictions for the high-redshift black hole mass function observable by JWST.

1 Introduction: The Eddington Bottleneck

Standard astrophysical models assume SMBHs grow from the remnants of Population III stars ($M_{seed} \sim 100 M_\odot$). The growth rate $\dot{M}$ is limited by the Eddington luminosity L_{Edd} , where outward radiation pressure balances inward gravitational pull. The mass growth is exponential:

$$M(t) = M_{seed} \exp\left(\frac{1 - \epsilon}{\epsilon} \frac{t}{t_{Edd}}\right) \quad (1)$$

where $\epsilon \approx 0.1$ is the radiative efficiency and $t_{Edd} \approx 0.45$ Gyr. For a quasar at $z = 7$ (age of universe $t \approx 0.77$ Gyr), growing from $100 M_\odot$ to $10^9 M_\odot$ requires roughly 16 e-folding times. However, the available time allows for only $\sim 10 - 12$ e-foldings, even assuming 100% duty cycles. This discrepancy forces standard theory to invoke continuous, unstable super-Eddington accretion. The APH framework resolves this by modifying the initial condition M_{seed} via vacuum phase transition dynamics.

2 Fractal Dust: The Remnants of Filtration

In the APH cosmology (1), the Big Bang is defined as the filtration of Sedenion zero divisors $\mathcal{Z}(\mathbb{S})$. This process is analogous to a rapid crystallization, which inevitably produces topological defects.

2.1 The Hausdorff Dimension of Defects

The set of zero divisors $\mathcal{Z}(\mathbb{S})$ is embedded in the 16-dimensional bulk but constrained by the isometry to G_2 (dim 14) and the fibration over $V_2(\mathbb{R}^7)$ (3). The granularity of the surviving defects is determined by the codimension of the filtration failure. We propose that the primordial *Sedenion Dust* forms a fractal set with dimension D_F :

$$D_F \approx \dim(\mathbb{S}) - \dim(G_2) = 16 - 14 = 2 \quad (2)$$

This implies that the primordial seeds originated as codimension-2 topological defects—filamentary structures or *Cosmic Sheets*—that underwent gravitational fragmentation immediately post-inflation.

2.2 Derivation of the Seed IMF

The mass distribution of these fragments is governed by the fractal scaling. The number density $n(M)$ of seeds follows a power law derived from the Press-Schechter formalism adapted for fractal density fields:

$$\frac{dN}{dM} \propto M^{-\alpha}, \quad \text{where } \alpha \approx \frac{D_F}{3} + 1 = \frac{5}{3} \quad (3)$$

Result: The APH Initial Mass Function (IMF) is top-heavy compared to the stellar Salpeter IMF ($\alpha = 2.35$). This naturally populates the early universe with rare, ultra-massive seeds ($10^4 - 10^6 M_\odot$) without requiring prior star formation.

3 Super-Eddington Accretion via Geometric Stiffness

The growth of these Sedenion seeds is further enhanced by their unique thermodynamic properties derived in (2).

3.1 The Entropy Sink Mechanism

A standard black hole has a horizon that acts as a one-way membrane, but the accretion flow is often choked by the heating of the gas outside the horizon. A **Sedenion Star**, however, is a material body with stiffness $\Gamma = \pi$ and internal convective degrees of freedom.

1. **No Hard Surface:** The core does not possess a solid crust; it is a hyper-stiff fluid. Infalling matter shocks not against a surface, but against the stiffness gradient.
2. **Algebraic Convection:** As derived in the Rice Convection Model for Kerr-Sedenion metrics (2), the core circulates zero-divisor flux tubes. This convection acts as an efficient transporter of entropy from the surface to the deep interior.

We model the effective radiative efficiency ϵ_{eff} for a Sedenion seed as:

$$\epsilon_{eff} \approx \epsilon_{std} \left(1 - \frac{L_{conv}}{L_{tot}} \right) \quad (4)$$

where L_{conv} is the power absorbed by internal algebraic convection. If $L_{conv} \rightarrow L_{tot}$, the seed becomes *black* (non-radiating) despite massive accretion. This bypasses the Eddington limit, allowing $\dot{M} \propto t^{-1}$ rather than exponential constraints.

4 Falsifiable Predictions

4.1 The High-Redshift Black Hole Mass Function

The fractal IMF predicts a specific excess of massive black holes at $z > 10$. **Observation:** JWST should detect Active Galactic Nuclei (AGN) with masses $M \sim 10^6 M_\odot$ residing in host galaxies with anomalously low stellar masses. **Ratio:** The standard ratio $M_{BH}/M_{bulge} \approx 10^{-3}$ should break down. APH predicts $M_{BH}/M_{bulge} \gg 10^{-3}$ (approaching unity) at $z > 10$, confirming the "Black Hole First" hierarchy.

4.2 No Population III Precursors

Standard theory predicts that high- z quasars should be chemically enriched by the supernovae of their Population III progenitors. **Observation:** APH Sedenion Seeds are *primordial vacuum condensates*, not stellar remnants. Their initial accretion disks should be pristine (metal-free) until galactic chemical evolution catches up. Spectroscopy of the highest-redshift quasars should reveal **metal-poor Broad Line Regions** despite the high black hole mass.

4.3 The Fractal Hum in the Gravitational Wave Background

The fragmentation of the fractal Sedenion sheets into discrete seeds ($z \sim 1000$) generates a stochastic gravitational wave background distinct from the 1.8 GHz *Scream*. **Observation:** We predict a secondary GW background in the **mHz to Hz** band (LISA/Decigo band), characterized by a spectral index $\Omega_{GW}(f) \propto f^{2/3}$, corresponding to the coalescence of dimension-2 fractal defects.

5 Conclusion

The *Impossible Quasars* are the direct descendants of the Big Bang's algebraic geometry. By identifying them as *Sedenion Seeds* formed from fractal vacuum dust, APH solves the time-constraint problem of the early universe. The black holes did not grow from stars; they were the anchors around which the first stars were born.

References

- [1] Schutz, A. M. (2025). *Pregeometric Cosmology: The Sedenion Scream*.
- [2] Schutz, A. M. (2025). *The Thermodynamics of Algebraic Collapse*.
- [3] Reggiani, S. (2024). *The Geometry of Sedenion Zero Divisors*. arXiv:2411.18881.